

TurtleBot3 개발 환경 구축



Contents

1 TurtleBot3

1.1 TurtleBot3라는 ROS를 배우기 위해 만들어진 작은 자율주행 모바일 로봇 플랫폼이 있다.

2 TurtleBot3 개발 환경 구축

2.1 TurtleBot3를 다루기 위해서 PC에 개발 환경을 구축해야한다.

2.2 TurtleBot3는 SLAM으로 Cartographer를 사용

2.3 혹시 Nav2 패키지가 설치 안 돼있다면

2.4 진정한 ROS 개발자라면 Colcon build 환경은 다 있죠? ^^

2.5 TurtleBot3 작업 공간을 만들고 이동

2.6 TurtleBot3 관련 소스를 내려받자

2.7 우리는 터틀봇을 시뮬레이션에서도 다루기 위해 Gazebo 관련 패키지도 미리 받자

2.8 혹시 모르니까 의존성 확인하고

2.9 이제 빌드

2.10 빌드했다고 바로 쓸 수 있는 게 아니죠~

2.11 TurtleBot3 모델 환경 설정

2.12 환경 변수 잘 설정됐는지 확인

2.13 공허한 세상에 터틀봇을 띄워보자

2.14 터미널 하나 더 열어서 터틀봇 전용 teleop keyboard 실행

2.15 홀로 곳곳이 살아가는 터틀봇

TurtleBot3 개발 환경 구축

Exported from Confluence · <https://pinkwink.atlassian.net>

TurtleBot3

TurtleBot3라는 ROS를 배우기 위해 만들어진 작은 자율주행 모바일 로봇 플랫폼이 있다.

Burger와 Waffle Pi 모델 2가지



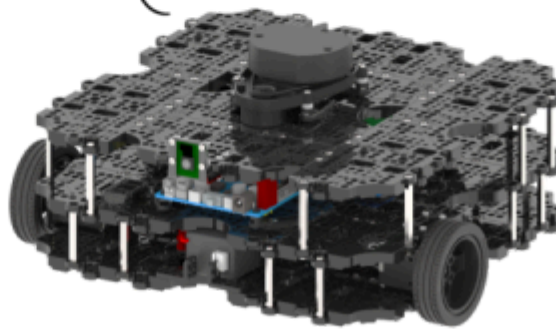
TURTLEBOT3



TurtleBot3
Burger



TurtleBot3
Waffle Pi



TurtleBot3 개발 환경 구축

TurtleBot3를 다루기 위해서 PC에 개발 환경을 구축해야한다.

ROBOTIS가 제공하는 e-Manual을 참조해서 진행 <https://emanual.robotis.com/docs/en/platform/turtlebot3/quick-start/> (<https://emanual.robotis.com/docs/en/platform/turtlebot3/quick-start/>)

Currently, ROS 1 Noetic and ROS 2 Humble are officially supported. In 2025, additional resources will be allocated for managing the open platform, with plans to complete example support for Humble in Q1 and extend support to Jazzy by Q2. The following chart provides an overview of the features supported by each ROS distribution.

Features	Noetic	Humble	Jazzy(soon)
Teleop	✓	✓	✓
SLAM	✓	✓	✓
Navigation	✓	✓	✓
Simulation	✓	✓	✓
Manipulation	✓	✓	X
Home Service Challenge	✓	X	X
Autonomous Driving	✓	✓	X
Machine Learning	X	X	X

3. Quick Start Guide



TurtleBot3 78 ROS 2 Humble Quick Start Guide
ROBOTIS OpenSourceTeam

TURTLEBOT3

Quick Start Guide

다음에서 보기: YouTube

TurtleBot3는 SLAM으로 Cartographer를 사용

Code

```
1 sudo apt install ros-jazzy-cartographer -y
2 sudo apt install ros-jazzy-cartographer-ros -y
```

```
colcon build [15/15 done] [0 ongoing]

ham@pc:~$ sudo apt install ros-jazzy-cartographer -y
sudo apt install ros-jazzy-cartographer-ros -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ros-jazzy-cartographer is already the newest version (2.0.9004-1noble.20260225.051215).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ros-jazzy-cartographer-ros is already the newest version (2.0.9003-2noble.20260412.051806).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ham@pc:~$
```

혹시 Nav2 패키지가 설치 안 되었다면

Code

```
1 sudo apt install ros-jazzy-navigation2 -y
2 sudo apt install ros-jazzy-nav2-bringup -y
```

colcon build [15/15 done] [0 ongoing]

```
ham@pc:~$ sudo apt install ros-jazzy-navigation2 -y
sudo apt install ros-jazzy-nav2-bringup -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ros-jazzy-navigation2 is already the newest version (1.3.11-1noble.20260412.073445).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ros-jazzy-nav2-bringup is already the newest version (1.3.11-1noble.20260412.073537).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ham@pc:~$
```

진정한 ROS 개발자라면 Colcon build 환경은 다 있죠? ^^

Code

```
1 sudo apt install python3-colcon-common-extensions
```

colcon build [15/15 done] [0 ongoing]

```
ham@pc:~$ sudo apt install python3-colcon-common-extensions
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-colcon-common-extensions is already the newest version (0.3.0-100).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ham@pc:~$
```

TurtleBot3 작업 공간을 만들고 이동

Code

```
1 mkdir -p ~/turtlebot3_ws/src
2 cd ~/turtlebot3_ws/src/
```

/bin/bash

```
ham@pc:~$ mkdir -p ~/turtlebot3_ws/src
ham@pc:~$ cd ~/turtlebot3_ws/src/
ham@pc:~/turtlebot3_ws/src$
```

TurtleBot3 관련 소스를 내려받자

Code

```
1 git clone -b jazzy https://github.com/ROBOTIS-GIT/DynamixelSDK.git
2 git clone -b jazzy https://github.com/ROBOTIS-GIT/turtlebot3_msgs.git
3 git clone -b jazzy https://github.com/ROBOTIS-GIT/turtlebot3.git
```

```
/bin/bash
ham@pc:~/turtlebot3_ws/src$ git clone -b jazzy https://github.com/ROBOTIS-GIT/DynamixelSDK.git
git clone -b jazzy https://github.com/ROBOTIS-GIT/turtlebot3_msgs.git
git clone -b jazzy https://github.com/ROBOTIS-GIT/turtlebot3.git
Cloning into 'DynamixelSDK'...
remote: Enumerating objects: 14874, done.
remote: Counting objects: 100% (2752/2752), done.
remote: Compressing objects: 100% (231/231), done.
remote: Total 14874 (delta 2639), reused 2521 (delta 2521), pack-reused 12122 (from 3)
Receiving objects: 100% (14874/14874), 28.38 MiB | 23.51 MiB/s, done.
Resolving deltas: 100% (8663/8663), done.
Cloning into 'turtlebot3_msgs'...
remote: Enumerating objects: 377, done.
remote: Counting objects: 100% (80/80), done.
remote: Compressing objects: 100% (45/45), done.
remote: Total 377 (delta 48), reused 45 (delta 33), pack-reused 297 (from 1)
Receiving objects: 100% (377/377), 87.88 KiB | 7.99 MiB/s, done.
Resolving deltas: 100% (175/175), done.
Cloning into 'turtlebot3'...
remote: Enumerating objects: 7981, done.
remote: Receiving objects: 87% (6944/7981), 65.18 MiB | 21.60 MiB/s
```

우리는 터틀봇을 시뮬레이션에서도 다루기 위해 Gazebo 관련 패키지도 미리 받자

Code

```
1 git clone -b jazzy https://github.com/ROBOTIS-GIT/turtlebot3_simulations.git
```

```
/bin/bash
ham@pc:~/turtlebot3_ws/src$ git clone -b jazzy https://github.com/ROBOTIS-GIT/turtlebot3_simulations.git
Cloning into 'turtlebot3_simulations'...
remote: Enumerating objects: 4365, done.
remote: Counting objects: 100% (1615/1615), done.
remote: Compressing objects: 23% (82/354)
```

혹시 모르니까 의존성 확인하고

명령어 실행 위치 주의

Code

```
1 cd ~/turtlebot3_ws
2 rosdep update
3 rosdep install --from-paths src --ignore-src -r -y
```

```
/bin/bash
ham@pc:~/turtlebot3_ws/src$ cd ~/turtlebot3_ws
rosdep update
rosdep install --from-paths src --ignore-src -r -y
/usr/bin/rosdep:6: DeprecationWarning: pkg_resources is deprecated as an API. See https://s
etuptools.pypa.io/en/latest/pkg_resources.html
  from pkg_resources import load_entry_point
reading in sources list data from /etc/ros/rosdep/sources.list.d
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/osx-homebrew.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/base.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/python.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/ruby.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/releases/fuerte.yaml
Query rosdistro index https://raw.githubusercontent.com/ros/rosdistro/master/index-v4.yaml
Skip end-of-life distro "ardent"
Skip end-of-life distro "bouncy"
Skip end-of-life distro "crystal"
Skip end-of-life distro "dashing"
Skip end-of-life distro "eloquent"
Skip end-of-life distro "foxy"
Skip end-of-life distro "galactic"
Skip end-of-life distro "groovy"
Add distro "humble"
```

이제 빌드

명령어 실행 위치 다시 한 번 더 주의!

Code

```
1 cd ~/turtlebot3_ws
2 colcon build --symlink-install
```



```
colcon build [5/15 done] [3 ongoing]

ham@pc:~/turtlebot3_ws$ colcon build --symlink-install
Starting >>> turtlebot3_msgs
Starting >>> dynamixel_sdk
Starting >>> turtlebot3_description
Starting >>> dynamixel_sdk_custom_interfaces
Starting >>> turtlebot3_cartographer
Starting >>> turtlebot3_gazebo
Starting >>> turtlebot3_navigation2
Starting >>> turtlebot3_teleop
Finished <<< turtlebot3_cartographer [1.31s]
Finished <<< turtlebot3_navigation2 [1.47s]
Finished <<< turtlebot3_description [1.92s]
Finished <<< turtlebot3_teleop [2.17s]
Finished <<< dynamixel_sdk [4.82s]
[5.9s] [5/15 complete] [3 ongoing] [turtlebot3_msgs:build 4% - 5.6s] ...
```

빌드했다고 바로 쓸 수 있는 게 아니죠~

빌드 결과를 매 터미널에서 자동으로 불러오도록 설정

```
Code

1 echo 'source ~/turtlebot3_ws/install/setup.bash' >> ~/.bashrc
2 source ~/.bashrc
```

```
colcon build [15/15 done] [0 ongoing]

ham@pc:~/turtlebot3_ws$ echo 'source ~/turtlebot3_ws/install/setup.bash' >> ~/.bashrc
ham@pc:~/turtlebot3_ws$ source ~/.bashrc
ham@pc:~/turtlebot3_ws$
```

TurtleBot3 모델 환경 설정

TurtleBot3는 버거와 와플 중 어떤 모델을 사용하든지 설정해야한다

Code

```
1 echo 'export TURTLEBOT3_MODEL=burger' >> ~/.bashrc
2 source ~/.bashrc
```

```
colcon build [15/15 done] [0 ongoing]
ham@pc:~/turtlebot3_ws$ echo 'export TURTLEBOT3_MODEL=burger' >> ~/.bashrc
ham@pc:~/turtlebot3_ws$ source ~/.bashrc
ham@pc:~/turtlebot3_ws$
```

환경 변수 잘 설정됐는지 확인

Code

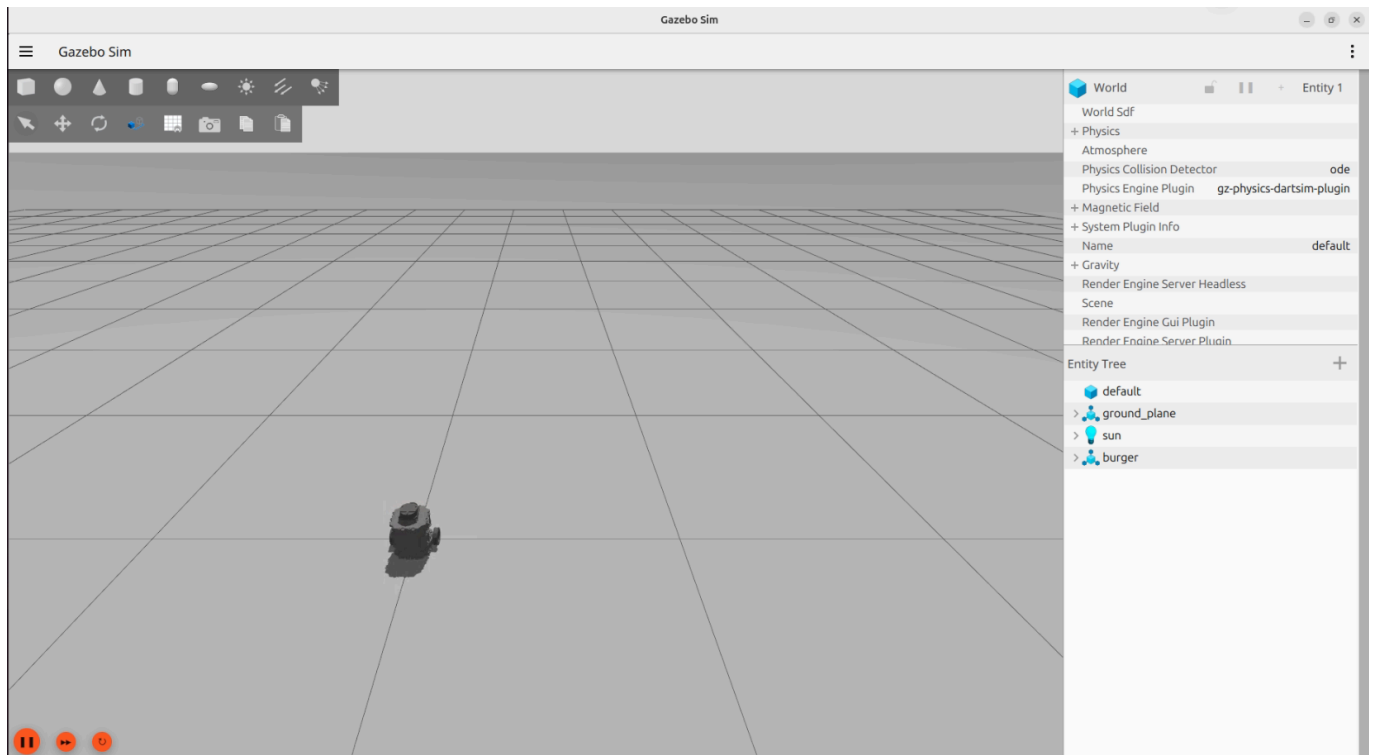
```
1 echo $TURTLEBOT3_MODEL
```

```
colcon build [15/15 done] [0 ongoing]
ham@pc:~/turtlebot3_ws$ echo $TURTLEBOT3_MODEL
burger
ham@pc:~/turtlebot3_ws$
```

공허한 세상에 터틀봇을 띄워보자

Code

```
1 ros2 launch turtlebot3_gazebo empty_world.launch.py
```



터미널 하나 더 열어서 터틀봇 전용 teleop keyboard 실행

터미널 하나 더 실행해서 터틀봇을 키보드로 조작할 수 있다면 준비 완료

Code

```
1  ros2 run turtlebot3_teleop teleop_keyboard
```

```
/bin/bash
ham@pc:~$ ros2 run turtlebot3_teleop teleop_keyboard

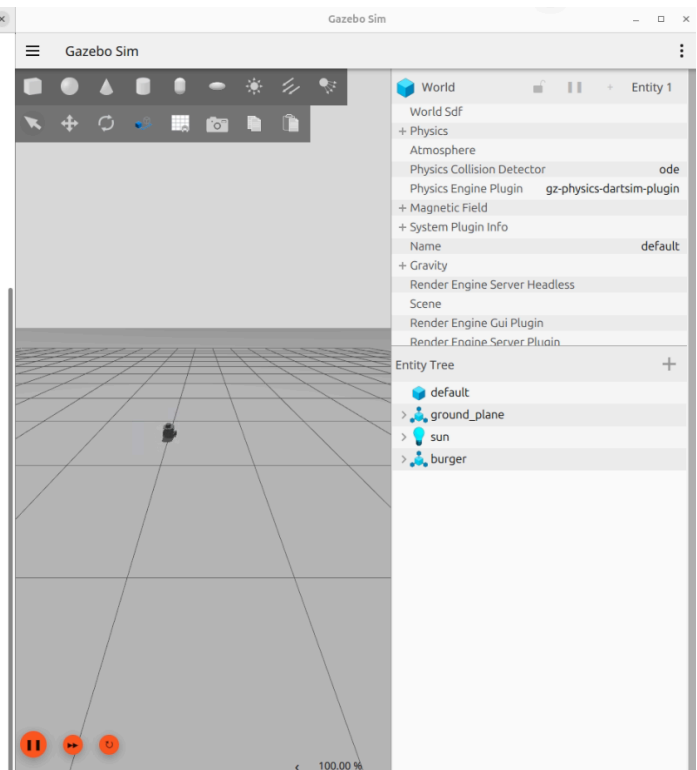
Control Your TurtleBot3!
-----
Moving around:
   w
 a   s   d
   x

w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle Pi : ~ 1.82)

space key, s : force stop

CTRL-C to quit
```

홀로 곳곳이 살아가는 터틀봇

[illegible]



PinkLAB YouTube Channel

PinkLAB Studio — Robotics, AI, ROS2 and more!

 **Subscribe on YouTube**

https://www.youtube.com/@pinklab_studio